

A Literature-Implied Partial Answer to Problem 3.19

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Status

Literature-implied answer, partial subcase. This note records an agent-identified implication of existing theorems. It should not be counted as an original proof. The supporting paper does not explicitly state that it answers Problem 3.19 of [1]; rather, its tensor-product modulation-space identifications combine with the Gabor coefficient theorem in [1] to give the sequence-space descriptions below.

Source Problem

The source paper is Debrouwere–Prangoski, *Gabor frame characterisations of generalised modulation spaces*, arXiv:2102.03217, published in *Analysis and Applications* [1]. In Section 3, Problem 3.19 asks:

$$\text{describe explicitly } (L^{p_1} \widehat{\otimes}_{\tau} L^{p_2})_d^{B_0}(\Lambda_1 \times \Lambda_2), \quad \tau \in \{\pi, \epsilon\},$$

for lattices $\Lambda_1, \Lambda_2 \subset \mathbb{R}^n$ and $1 \leq p_1, p_2 < \infty$.

Identified Subcase

Combining [1, Corollary 4.11] with [2, Corollary 6.1 and Remark 6.2] gives the following partial description of Problem 3.19.

- If $1 \leq p_1 \leq p_2 \leq 2$, then

$$(L^{p_1} \widehat{\otimes}_{\pi} L^{p_2})_d^{B_0}(\Lambda_1 \times \Lambda_2) \cong \ell^1(\Lambda_1; \ell^{p_2}(\Lambda_2))$$

topologically.

- If $2 \leq p_2 \leq p_1 < \infty$, then

$$(L^{p_1} \widehat{\otimes}_{\epsilon} L^{p_2})_d^{B_0}(\Lambda_1 \times \Lambda_2) \cong c_0(\Lambda_1; \ell^{p_2}(\Lambda_2))$$

topologically.

These formulae extend the special $p_2 = 2$ descriptions stated in [1, Corollary 3.21]; the projective formula also contains the source paper’s explicitly mentioned trivial case $p_1 = 1$.

Reason For The Identification

Let $F = L^{p_1} \widehat{\otimes}_\pi L^{p_2}$. The supporting tensor-product paper [2, Corollary 6.1(a) and Remark 6.2] proves

$$\mathcal{M}[F] = W(\mathcal{F}L^{p_2}, L^1) = \mathcal{F}M^{p_2,1}, \quad 1 \leq p_1 \leq p_2 \leq 2.$$

The source paper’s Gabor coefficient theorem [1, Corollary 4.11] identifies $\mathcal{M}[F]$, through the coefficient map associated with a suitable dual Gabor pair, with F_d^σ , where $F_d^\sigma = F_d^{B_0}$. The standard Gabor coefficient description of $\mathcal{F}M^{p_2,1} = W(\mathcal{F}L^{p_2}, L^1)$ is $\ell^1(\Lambda_1; \ell^{p_2}(\Lambda_2))$. This yields the projective formula.

Similarly, for $F = L^{p_1} \widehat{\otimes}_\epsilon L^{p_2}$, [2, Corollary 6.1(b)] proves

$$\mathcal{M}[F] = W(\mathcal{F}L^{p_2}, L_0^\infty), \quad 2 \leq p_2 \leq p_1 < \infty.$$

Applying [1, Corollary 4.11] and the standard Gabor coefficient description of $W(\mathcal{F}L^{p_2}, L_0^\infty)$ gives $c_0(\Lambda_1; \ell^{p_2}(\Lambda_2))$.

For arbitrary product lattices, one may pass to a sufficiently fine finite-index product refinement that admits the required smooth dual Gabor pair, apply the preceding identification there, and restrict by zero extension to the original sub-lattice. Since the definition of E_d^B is independent of the compact test function used in [1, Theorem 3.1], this restriction preserves the corresponding mixed sequence-space description.

Scope Limitations

This does not settle Problem 3.19 for all exponents. The remaining regions include the projective tensor product outside $1 \leq p_1 \leq p_2 \leq 2$ and the injective tensor product outside $2 \leq p_2 \leq p_1 < \infty$. The supporting authors did not explicitly advertise their result as an answer to Problem 3.19; the relation is a reformulation through the source paper’s Gabor coefficient theorem.

Verification Notes

The cheap run indexes were searched for arXiv:2102.03217, the title, Gabor frames, generalized modulation spaces, and tensor-product L^p discrete spaces. No previous packet for this source paper was found. Web and local source searches found the published source metadata and the related Feichtinger–Pilipovic–Prangoski tensor-product modulation paper, but no separate explicit statement of the above B_0 -discrete-space answer.

References

- [1] A. DeBrouwere and B. Prangoski, *Gabor frame characterisations of generalised modulation spaces*, Analysis and Applications 21 (2023), no. 3, 547–596. arXiv:2102.03217. DOI: 10.1142/S0219530522500178.
- [2] H. G. Feichtinger, S. Pilipovic, and B. Prangoski, *Modulation spaces associated with tensor products of amalgam spaces*, Annali di Matematica Pura ed Applicata 201 (2022), 127–155. arXiv:2012.12295. DOI: 10.1007/s10231-021-01110-9.